Secure Personal Health Vault

Three-Algorithm Performance Evaluation

1. Scope and objective

This report compares three configurations: AES-256-GCM, DES-CBC, and RC4. AES-GCM is the recommended choice for protecting real vault data because it provides authenticated encryption. DES and RC4 are obsolete and are included only for academic comparison; they must not be used to protect real personal or health records.

The benchmark measures the implementation configurations available in this project. AES-GCM includes its authentication tag. DES-CBC and RC4 use separate HMAC-SHA256 integrity protection in the application, so raw-cipher timing is not a comparison of identical security mechanisms. Application download timings include more than the cipher operation.

2. Environment and methodology

Environment item	Value
Python	3.13.7 (CPython)
Cryptography library	50.0.2
OpenSSL	OpenSSL 4.0.3 29 Sep 2026
Operating system	Windows-11-10.0.26200-SP0
Processor	Intel64 Family 6 Model 183 Stepping 1, GenuineIntel
Logical CPUs visible	24
Measured trials	30
Warm-up runs	3
Benchmark started (UTC)	2026-10-27T12:41:12Z
Benchmark finished (UTC)	2026-10-27T12:43:30Z

The benchmark uses synthetic input files and measures encryption and decryption separately. Throughput is reported in MiB/s, where 1 MiB = 1,048,576 bytes. Timing results are machine- and workload-dependent; they should not be generalized to every computer or deployment.

3. Mean performance on 10 MiB input

Algorithm	Operation	Mean (ms)	SD (ms)	Mean throughput (MiB/s)	95% CI half-width (ms)
AES-256-GCM	encrypt	5.669	0.771	1763.88	0.276
AES-256-GCM	decrypt	5.992	0.779	1668.78	0.279
DES-CBC (legacy)	encrypt	253.871	9.985	39.39	3.573
DES-CBC (legacy)	decrypt	246.280	9.414	40.60	3.369

Algorithm	Operation	Mean (ms)	SD (ms)	Mean throughput (MiB/s)	95% CI half-width (ms)
RC4 (legacy stream cipher)	encrypt	19.745	1.120	506.47	0.401
RC4 (legacy stream cipher)	decrypt	20.056	1.051	498.61	0.376

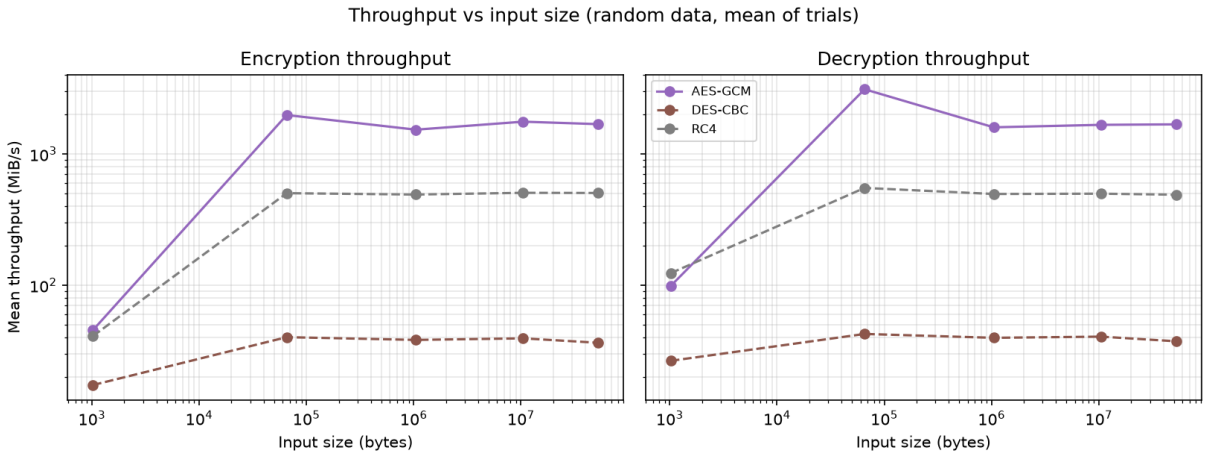
4. Throughput across file sizes

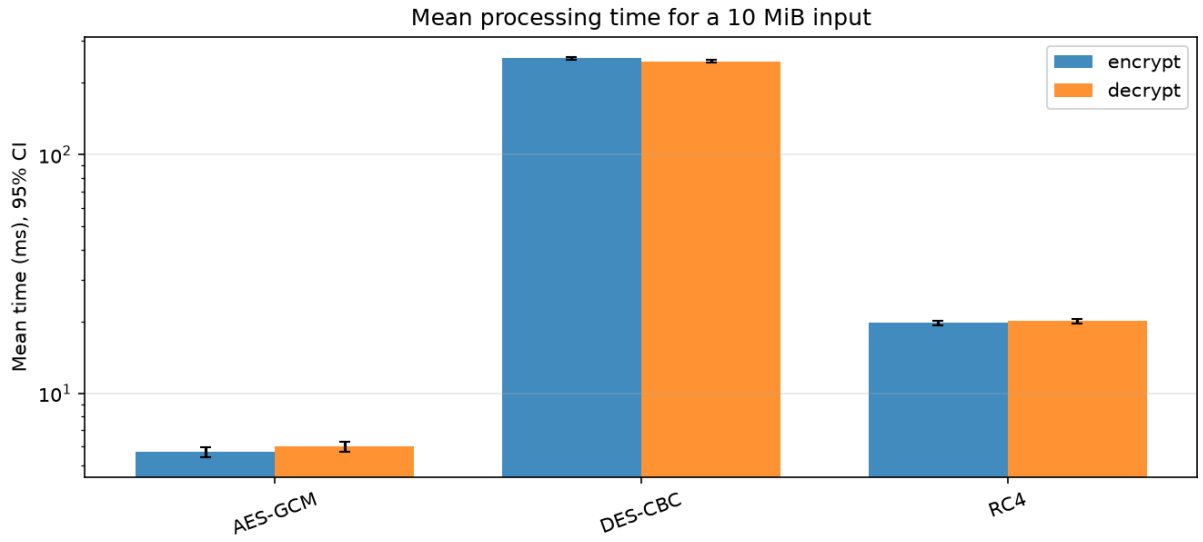
Encrypt throughput

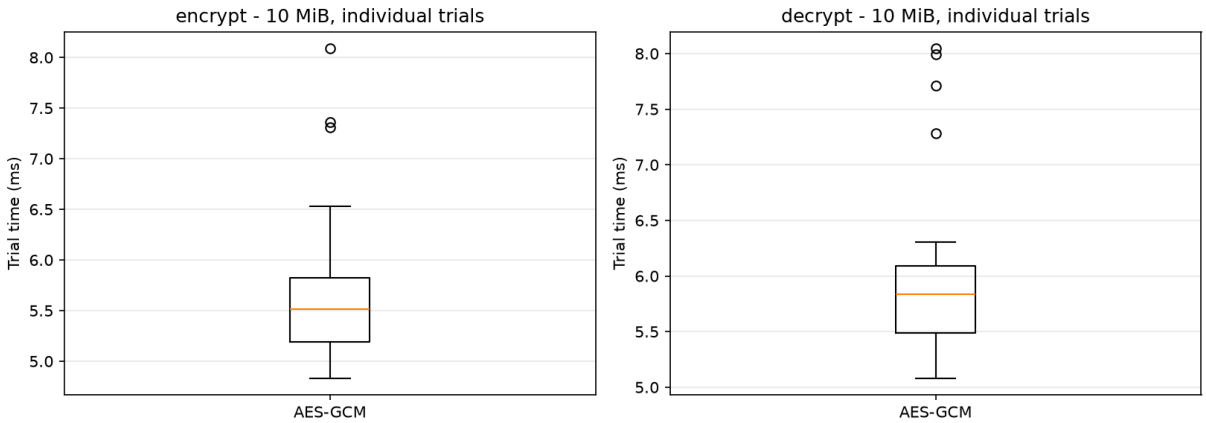
Algorithm	1 KiB	64 KiB	1 MiB	10 MiB	50 MiB
AES-256-GCM	45.53	1981.82	1532.86	1763.88	1689.33
DES-CBC (legacy)	17.37	40.22	38.34	39.39	36.46
RC4 (legacy stream cipher)	41.03	503.14	491.05	506.47	504.93

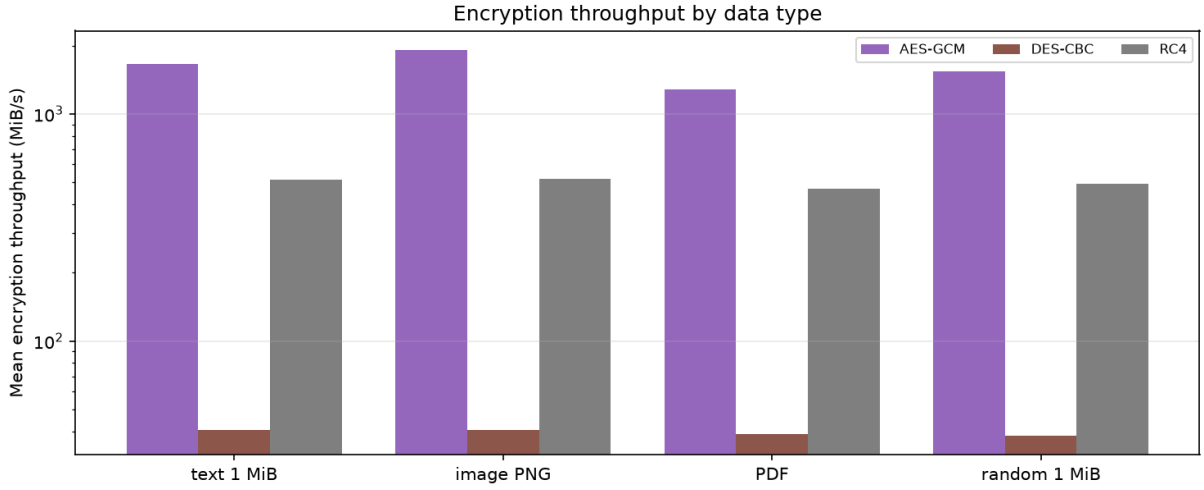
Decrypt throughput

Algorithm	1 KiB	64 KiB	1 MiB	10 MiB	50 MiB
AES-256-GCM	98.98	3115.14	1597.84	1668.78	1682.97
DES-CBC (legacy)	26.57	42.56	39.77	40.60	37.42
RC4 (legacy stream cipher)	123.77	551.58	496.08	498.61	489.81







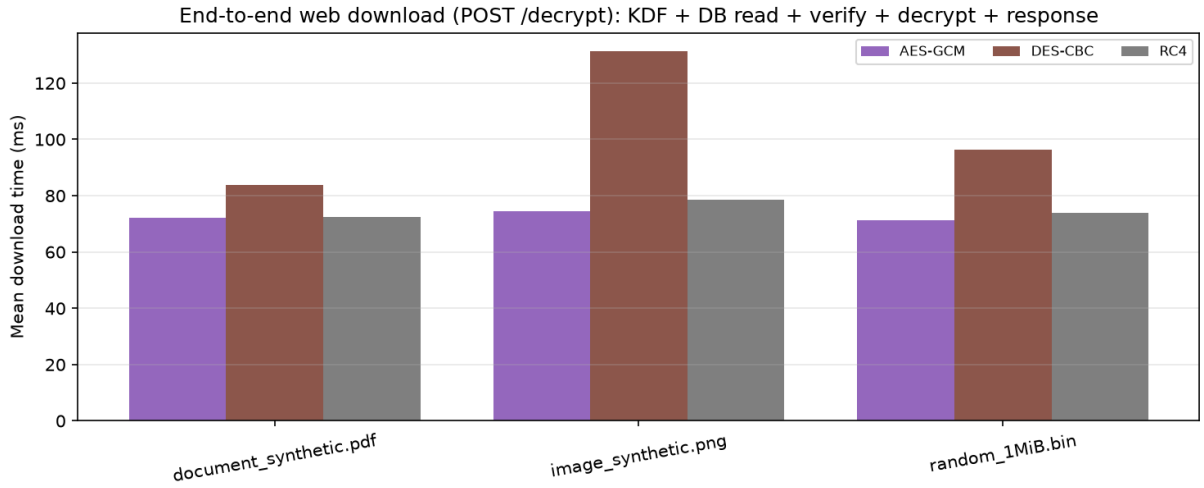


5. End-to-end application download

Algorithm	document_synthetic.pdf	image_synthetic.png	random_1MiB.bin
AES-256-GCM	72.23	74.43	71.31
DES-CBC (legacy)	83.78	131.33	96.24

Algorithm	document_synthetic.pdf	image_synthetic.png	random_1MiB.bin
RC4 (legacy stream cipher)	72.42	78.56	73.90

These timings include application-level work such as key derivation, database access, decryption, and response handling, depending on the benchmark harness. They should not be interpreted as raw cipher throughput.



6. File I/O measurements

File	Size (bytes)	Mean read (ms)	Mean write (ms)
random_1KiB.bin	1024	0.087	0.381
random_64KiB.bin	65536	0.150	0.495
document_synthetic.pdf	559953	0.183	1.572
text_1MiB.txt	1048576	0.299	0.808
random_1MiB.bin	1048576	0.355	0.698
image_synthetic.png	2457390	0.870	1.015
random_10MiB.bin	10485760	3.050	3.173
random_50MiB.bin	52428800	14.880	28.133

7. Integrity and ciphertext observations

Algorithm	Result after ciphertext bit flip	Plaintext rejected
AES-256-GCM	decryption rejected (authentication/padding failure)	Yes
DES-CBC (legacy)	decrypted with errors	No
RC4 (legacy stream cipher)	decrypted with errors	No

Ciphertext sizes

Algorithm	Plaintext 0 B	Plaintext 1 B	Plaintext 15 B	Plaintext 16 B	Plaintext 17 B	Plaintext 31 B	Plaintext 32 B	Plaintext 1000 B
AES-256-GCM	16	17	31	32	33	47	48	1016
DES-CBC (legacy)	8	8	16	24	24	32	40	1008
RC4 (legacy stream cipher)	0	1	15	16	17	31	32	1000

8. Interpretation and limitations

• AES-GCM is the recommended choice for the vault because it provides confidentiality and integrity using authenticated encryption. Its 12-byte nonce must never be reused with the same key; the 16-byte authentication tag detects tampering.

• DES is obsolete and its effective key strength is inadequate for modern security. RC4 is an obsolete stream cipher with known keystream biases. Both are included only for academic comparison.

• The app uses a separate HMAC-SHA256 for DES-CBC and RC4, whereas AES-GCM has a built-in authentication tag. Raw benchmark timings do not include all application security overhead unless the benchmark explicitly measures the full application path.

• Results depend on CPU, operating system, cryptography/OpenSSL versions, input size, system load, and number of trials. A single run is not a universal performance ranking.

• Synthetic fixtures are not real patient data. The benchmark does not establish legal compliance or production readiness.

• Separate KDF timing data: PBKDF2 iteration count 600000; mean 101.41 ms across 20 trials. This value is contextual and is not added to raw cipher-only timings.

9. Conclusion

The benchmark compares AES-256-GCM, DES-CBC, and RC4 using the available measurements. Select AES-GCM for real vault data; the legacy algorithms are retained only to satisfy the academic comparison requirement. Discuss measured differences using the generated tables and graphs, while distinguishing raw cipher timing from application download latency.

Generated from the current files in `benchmark/results/`. Regenerate this report after the final benchmark run; do not submit filtered or stale results as fresh measurements.